

Cyber Security Skill Development Lab

(2026-28)

Lab Record

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Section / Batch	CyberSecurity (2026-28)

Lab Assignment 1

Windows Operating System Fundamentals and Security

1. Aim / Objective

Aim: To develop a practical understanding of Windows operating-system fundamentals and security mechanisms by working with a Windows 10 virtual machine.

Objectives:

1. To understand the major stages involved in the Windows boot process.
2. To understand the difference between authentication and authorization.
3. To create and manage local Windows user accounts and groups.
4. To understand the difference between administrator and standard-user privileges.
5. To study the purpose and working of User Account Control (UAC).
6. To use Command Prompt and PowerShell for basic system administration and security analysis.
7. To understand how file and folder permissions control access to resources.
8. To study the security risks associated with excessive privileges and inappropriate permissions.
9. To identify common security weaknesses related to Windows user and access management.

2. System/VM configuration

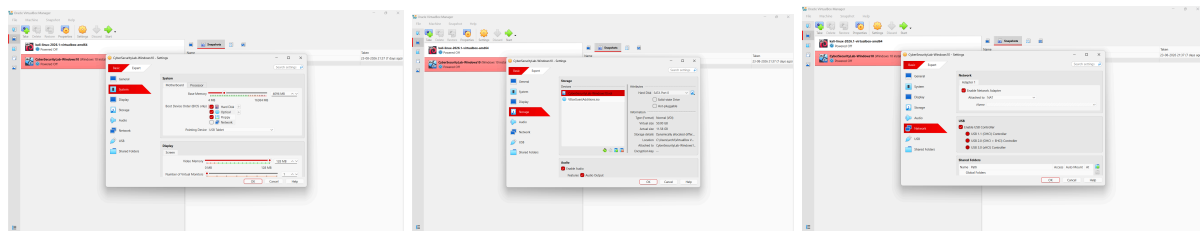
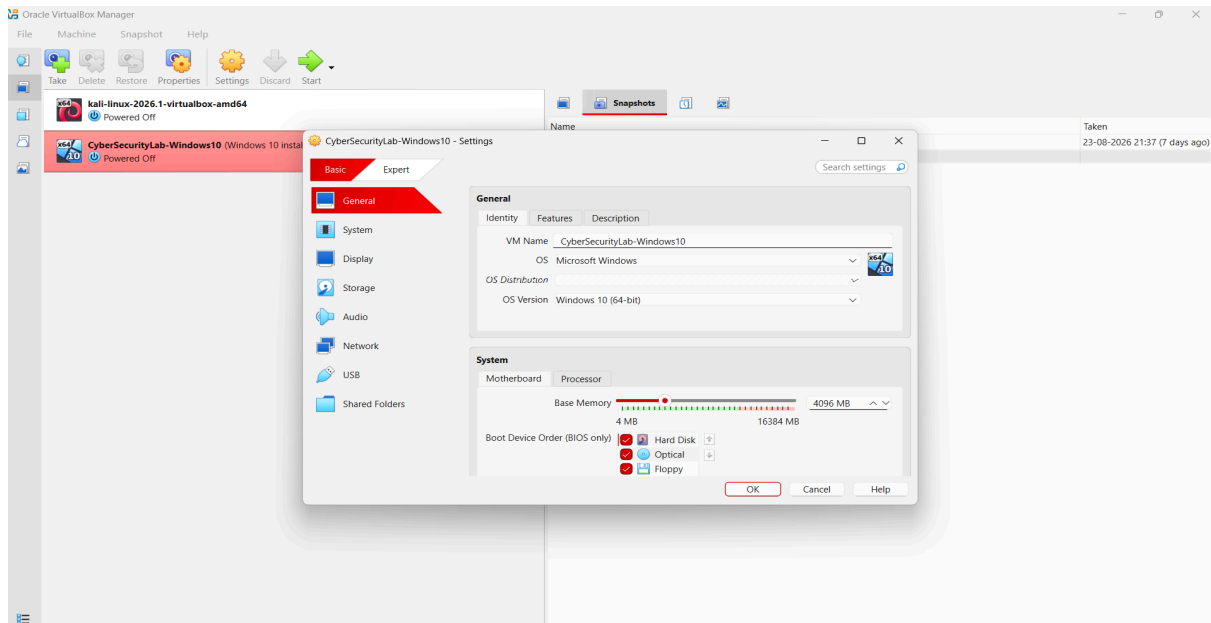
The experiment was performed inside a **Windows 10 virtual machine** using **Oracle VM VirtualBox**. A virtual machine was used so that the different user, permission, access-control and security experiments could be performed in a controlled laboratory environment without affecting the main host operating system.

Before starting the experiment, a **snapshot of the VM** is created to save its current working state. A snapshot acts as a **restore point**, allowing the user to return to the previous state if any changes cause problems. Since the experiment may involve modifying **users, permissions, or system configurations**, there is a risk of affecting the system. If something goes wrong, the VM can be restored from the snapshot without manually undoing all the changes. This makes it safer to perform security-related experiments.

Virtual Machine Configuration:

Parameter	Configuration
Virtualization Software	Oracle VM VirtualBox
Virtual Machine Name	CyberSecurityLab-Windows10
Guest Operating System	Windows 10
System Architecture	64-bit
Virtual Processors	2 CPU cores
Allocated RAM	4 GB
Virtual Disk	50 GB
Network Configuration	NAT
Shell	Cmd & PowerShell

The configuration follows the recommended VM configuration given in the assignment- 2 virtual CPU cores, 4 GB or more RAM, a 40–60 GB virtual disk and NAT networking initially.



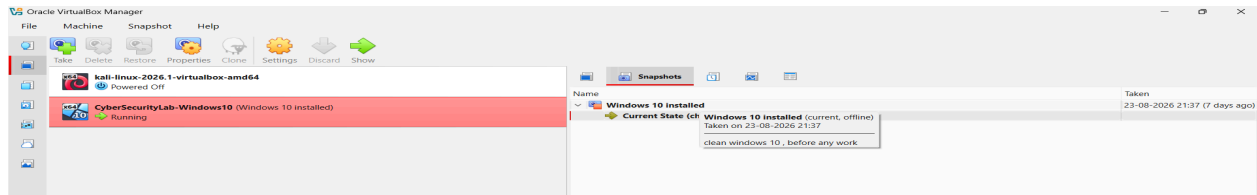
Settings dialog for the VM, showing the System, Processor and Network tabs

3. Installation procedure

The VM was built and prepared in the following order.

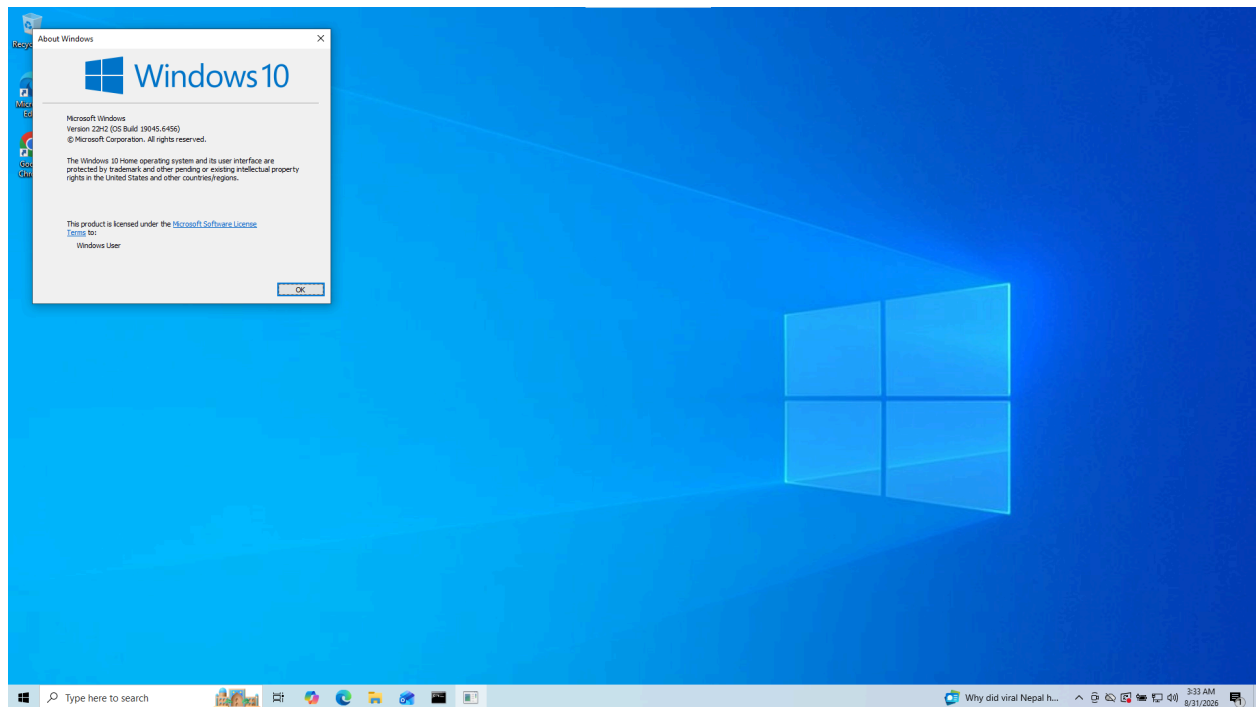
1. Downloaded and installed Oracle VM VirtualBox on the host machine, then verified the installation by launching VirtualBox Manager.
2. Used the “New Virtual Machine” wizard to create a new machine: named it CyberSecurityLab-Windows10, set Type to Microsoft Windows and Version to Windows 10 (64-bit).
3. Allocated 4096 MB of RAM and 2 virtual CPUs, then created a new virtual hard disk (VDI, dynamically allocated, 50 GB).
4. Attached the official Windows 10 ISO image to the VM's virtual optical drive and started the VM.
5. Completed Windows Setup: selected language and edition, chose Custom install onto the virtual disk, and created a local account for use inside the lab environment.
6. Once the desktop loaded cleanly, opened Machine , Took Snapshot in VirtualBox Manager and saved a snapshot named “Windows10 installed”, so that the VM could be restored to a clean state if a later experiment left the system in an unusable condition.

- Created Snapshot of base windows10



Snapshots pane

- Verified windows 10 installation by running winver command:



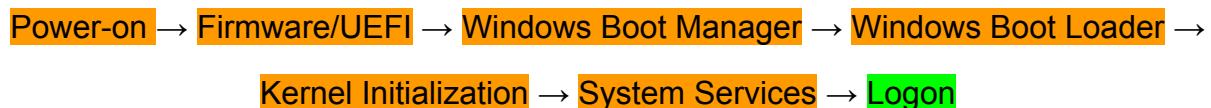
Windows 10 desktop after first successful login

4. Windows boot-sequence

Windows 10 boots in a fixed, well-defined sequence, moving from firmware-level hardware initialisation through progressively higher-level software until a user session is established.

Stages of Windows Boot Sequence

Windows boot Sequence Diagram:



1. Power-On :

- When the system is powered on, the CPU begins executing the startup instructions stored in the system firmware.

2. Firmware / UEFI :

- POST (Power-On Self-Test): When the computer starts, UEFI performs a basic hardware check before loading the operating system. It verifies that essential components such as RAM, CPU, keyboard, storage, and other required hardware are detected and functioning well enough to continue booting. The operating system cannot start properly if critical hardware is unavailable or malfunctioning. POST helps detect such problems at the earliest stage and may report errors before Windows loads.
- After POST, UEFI checks the configured boot devices and looks for a valid bootloader. In a modern Windows system, it normally looks for the EFI System Partition (ESP) on the selected storage device where UEFI boot files are stored. UEFI loads the boot manager from this partition and transfers control to it, allowing the Windows boot process to continue.
- UEFI is one of the earliest software components to execute. If an attacker compromises the firmware or replaces trusted boot files, they may gain control before Windows and its normal security mechanisms are loaded. This is one reason technologies such as Secure Boot are used to verify trusted boot components.

3. Windows Boot Manager ([bootmgfw.efi](#)) :

- Its main role is to decide how and which Windows installation should be started.
- Reads BCD (Boot Configuration Data): The Boot Manager reads the BCD store, which contains boot-related settings such as the Windows installation location, available operating systems, boot options, and recovery settings.
- Selects the boot entry: If multiple operating systems or boot options are available, the Boot Manager can display a boot menu and allow the user to select one. If there is only one normal Windows installation, it generally proceeds automatically.

The Boot Manager is security-sensitive because modifying the BCD or boot files can change the way Windows starts. For example, an attacker who gains the ability to modify boot configuration could attempt to disable or bypass certain startup security controls. Secure Boot helps protect this stage by allowing only trusted, digitally signed boot components to execute.

4. Windows Boot Loader ([winload.efi](#)):

- Its main job is to prepare Windows for the kernel to take control. It locates and loads the Windows kernel (`ntoskrnl.exe`) into memory. It also loads the Hardware Abstraction Layer (HAL) and essential boot-start drivers required for Windows to communicate with hardware such as storage devices.

5. Kernel Initialisation :

- It acts as the intermediary between applications, hardware, memory, and system resources. It initializes important components such as memory management, process and thread management, hardware, interrupt handling, and additional device drivers. It establishes the basic environment required for Windows to operate and starts essential system processes. The kernel also begins enforcing important access-control and security mechanisms at the operating-system level.
6. System Services :
- Once the kernel has initialized the core operating-system environment, Windows starts system processes, drivers, and essential services required for normal operation.
 - The Service Control Manager (SCM) manages the starting, stopping, and configuration of Windows services.
 - Services provide functionality such as networking, event logging, security, hardware management, and other background operations.
7. Logon:
- After the required system services are running, Windows reaches the **logon stage**, where it is ready for a user to access the system.
 - **Winlogon (winlogon.exe)** manages the Windows interactive logon process. It displays the appropriate sign-in interface and coordinates with Windows authentication components

bcdedit Output :

bcdedit is a Windows utility used to view and modify the Boot Configuration Data (BCD) that controls the Windows boot process.

```

Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>bcdedit

Windows Boot Manager
-----
identifier           {bootmgr}
device               partition=C:
description           Windows Boot Manager
locale               en-US
inherit              {globalsettings}
default              {current}
resumeobject         {bfc027ef-9f6d-11f1-ad70-be98c7b85cc5}
displayorder         {current}
toolsdisplayorder    {memdiag}
timeout              30

Windows Boot Loader
-----
identifier           {current}
device               partition=C:
path                 \Windows\system32\winload.exe
description           Windows 10
locale               en-US
inherit              {bootloadersettings}
recoverysequence     {bfc027f1-9f6d-11f1-ad70-be98c7b85cc5}
displaymessageoverride Recovery
recoveryenabled       Yes
allowedinmemorysettings 0x15000075
osdevice             partition=C:
systemroot            \Windows
resumeobject         {bfc027ef-9f6d-11f1-ad70-be98c7b85cc5}
nx                   OptIn
bootmenupolicy        Standard

C:\Windows\system32>

```

bcdedit command executed from an Administrator Command Prompt.

1. Windows Boot Manager

The first section displays the configuration of the Windows Boot Manager:

- **identifier {bootmgr}** — Identifies this entry as the Windows Boot Manager.
- **device partition=C:** — Indicates that the boot manager configuration is associated with the C: partition in this configuration.
- **description Windows Boot Manager** — Name of the boot manager entry.
- **locale en-US** — Specifies the language/locale used for the boot environment.
- **default {current}** — The current Windows installation is configured as the default boot entry.
- **displayorder {current}** — Specifies the order in which boot entries are presented.
- **timeout 30** — The boot menu waits 30 seconds before automatically selecting the default entry.

2. Windows Boot Loader

The second section shows the configuration used to actually load Windows:

- `identifier {current}` — Refers to the Windows installation currently being used.
- `device partition=C:` — Windows is located on the C: partition.
- `path \Windows\system32\winload.exe` — Specifies the Windows Boot Loader that is executed to start Windows.
- `description Windows 10` — Name assigned to this Windows installation.
- `osdevice partition=C:` — Indicates that the operating system files are located on C:.
- `systemroot \Windows` — Specifies the Windows system directory.
- `recoveryenabled Yes` — Windows Recovery Environment is enabled for this installation.
- `bootmenupolicy Standard` — Uses the standard Windows boot menu interface.
- `nx OptIn` — Indicates the configured DEP/NX (No Execute) policy, which helps prevent execution of code from memory regions that should contain only data.

Security Relevance-

The screenshot demonstrates why the BCD is an important security component of the boot process. The BCD contains configuration information that determines which Windows installation is started, where it is located, which boot loader is used, recovery settings, and boot behavior. If an unauthorized user with sufficient privileges modifies these settings, they could potentially alter the boot process or weaken certain protections.

Effect of Failure at Different Stages

- Power-On / UEFI: Hardware or firmware failure may prevent the boot process from starting or prevent the system from detecting a bootable device.
- Windows Boot Manager: Failure or corruption of the Boot Manager or EFI System Partition (ESP) can prevent Windows from being located or started.
- BCD: Corruption or incorrect BCD configuration can cause incorrect boot selection or boot configuration errors.
- Windows Boot Loader: Failure of `winload.efi` or required boot drivers can prevent the kernel from being loaded and may cause Windows to enter Recovery Environment.
- Kernel Initialization: Kernel or critical driver failures can cause startup failure, system crashes, or a Blue Screen of Death (BSOD).
- System Services: Failure of essential services may allow Windows to start but prevent important system functions from operating correctly including security systems.
- Logon: Authentication or `Winlogon` failures may prevent the user from accessing their Windows session even though the operating system itself has started.

Overall, An earlier-stage failure generally has a greater impact on system availability, because each stage depends on the successful completion of the previous stage.

Security Concepts in the Boot Process

The boot process is closely related to system security because several critical components execute before the normal Windows environment and security software are fully active. Secure Boot, implemented through UEFI, verifies that trusted and digitally signed boot components are loaded, helping prevent unauthorized boot software from executing during startup.

- **Boot Integrity:** Ensures that important boot components have not been unauthorizedly modified, replaced, or corrupted. Security-sensitive components include UEFI firmware, EFI System Partition (ESP), Windows Boot Manager, BCD, and Windows Boot Loader.
- **Rootkit:** A type of malware designed to hide malicious activity and maintain privileged access to a system. It may manipulate system components, processes, or files to make its presence difficult to detect.
- **Bootkit:** A specialized type of rootkit that targets the boot process. It attempts to modify or interfere with boot components so that malicious code can execute very early during system startup, potentially before normal security software is running.
- **Security Significance:** If an attacker successfully modifies an early boot component, they may be able to establish persistent control and evade detection. Secure Boot helps reduce this risk by preventing untrusted or improperly signed boot components from being executed.

Overall , Secure Boot and boot integrity help establish a trusted chain of execution from UEFI through the Windows boot components, protecting the system against attacks that attempt to gain control during the early boot process

5. User and group configuration

For this lab, one administrator account, two standard-user accounts, and one custom local group were created, as required by Part C of the assignment.

A **user account** represents an individual identity on a Windows system. It is used to log in and has specific permissions and privileges while, A **group** is a collection of users that helps manage permissions more efficiently. Instead of assigning the same permissions to each user individually, permissions can be assigned to the group, and its members can receive those permissions through their group membership.

Accounts and Group Created

Name	Type	Purpose
LabAdmin	Administrator (member of Administrators)	Used for all elevation,Administrative tasks

Student1	Standard user (member of Users only)	Less privileges , can do limited tasks only
Student2	Standard user (member of Users only)	Less privileges , can do limited tasks only
CyberLabUsers	Custom local group	used to demonstrate group-based rather than per-user permission management.

Command-Line Method

The accounts and group were also created using Command Prompt

`net user LabAdmin * /add` - To create LabAdmin

`net localgroup Administrators AdminUser /add` - To add LabAdmin to administrators Group

`net user Student1 * /add` - To Create Student1 Account

`net user Student2 * /add` - To Create Student2 Account

`net localgroup LabUsers /add` - To add Labusers Group

`net localgroup LabUsers StdUser1 StdUser2 /add` - to add studnet1 and student2 to LabUsers Group

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>net user

User accounts for \\CYBERSECURITYLA

-----
Administrator      CSLab              DefaultAccount
Guest              LabAdmin           Student1
Student2           WDAUtilityAccount
The command completed successfully.

C:\Windows\system32>net localgroup cyberlabusers
Alias name     cyberlabusers
Comment
Members

-----
Student1
Student2
The command completed successfully.

C:\Windows\system32>net localgroup administrators
Alias name     administrators
Comment       Administrators have complete and unrestricted access to the computer/domain
Members

-----
Administrator
CSLab
LabAdmin
The command completed successfully.

C:\Windows\system32>
```

Commands Used-

Command	Purpose	Key Options	Security Relevance
<code>whoami</code>	Displays the identity of the user currently logged in, including the computer/domain name and username.	<code>/user</code> – displays the SID; <code>/groups</code> – displays group membership; <code>/priv</code> – displays privileges; <code>/all</code> – displays all available information.	This is usually one of the first commands used during an audit because it confirms which account is currently being used. This is important when checking whether actions are being performed with administrator or standard-user privileges.
<code>Whoami/groups</code>	Displays the groups to which the current user belongs, including built-in groups and security identities.	-	Helps verify whether a user has membership in a privileged group, such as Administrators. It can also reveal unexpected group memberships that may provide additional access.
<code>net user</code>	Lists all local user accounts present on the computer.	<code>/add</code> – creates an account; <code>/delete</code> – removes an account; <code>/active:{yes no}</code> – enables or disables an account.	Provides a basic account inventory. An unknown or unnecessary account may indicate poor account management or, in some situations, unauthorized access.
<code>net user <username></code>	Displays detailed information about a particular user account, such as its active status, password information, and group memberships.	<code>/domain</code> – queries domain information; <code>/times</code> – displays allowed logon times.	Helps verify whether an account is active, how its password is configured, and which groups it belongs to. This is useful when checking for excessive privileges or weak account management.
<code>net localgroup</code>	Lists all local groups configured on the computer.	-	Shows the groups available for controlling access. Groups such as Administrators, Remote Desktop Users, and Backup Operators can provide significant privileges, so reviewing them is useful during a security audit.

<code>net localgroup <groupname></code>	Displays the users who are members of a specific local group. For example, <code>net localgroup Administrators</code> shows members of the Administrators group.	<code>/add</code> – adds a user or group; <code>/delete</code> – removes a user or group.	Helps determine who has access through a particular group. Checking the Administrators group is especially important because its members have extensive control over the system.
--	---	--	---

```

Select Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\CSLab\whoami
cybersecurity\cs\slab

C:\Users\CSLab\whoami /groups

GROUP INFORMATION
-----
Group Name                                     Type                SID                Attributes
-----
Everyone                                     Well-known group    S-1-1-0            Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Local account and member of Administrators group  Well-known group    S-1-5-114          Group used for deny only
BUILTIN\Administrators                      Alias               S-1-5-32-544       Group used for deny only
BUILTIN\Users                              Alias               S-1-5-32-545       Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\INTERACTIVE                   Well-known group    S-1-5-4            Mandatory group, Enabled by default, Enabled group
CONSOLE LOGON                             Well-known group    S-1-2-1            Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Authenticated Users            Well-known group    S-1-5-11           Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\This Organization               Well-known group    S-1-5-15           Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Local account                  Well-known group    S-1-5-113          Mandatory group, Enabled by default, Enabled group
LOCAL                                       Well-known group    S-1-2-0            Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\NTLM Authentication             Well-known group    S-1-5-64-10        Mandatory group, Enabled by default, Enabled group
Mandatory Label\Medium Mandatory Level      Label               S-1-16-8192

C:\Users\CSLab>net user

User accounts for \\CYBERSECURITY\LA
-----
Administrator      CSLab      DefaultAccount
Guest               LabAdmin   Student1
Student2            NoGuilityAccount

The command completed successfully.

C:\Users\CSLab>net user LabAdmin
User name           LabAdmin
Full Name
Comment
User's comment
Country/region code 000 (System Default)
Account active       Yes
Account expires      Never

Password last set    8/31/2026 3:42:37 AM
Password expires     10/12/2026 3:42:37 AM
Password changeable  8/31/2026 3:42:37 AM
Password required    Yes
User may change password Yes

Workstations allowed All
Logon script
User profile
Home directory
Last logon           Never

Logon hours allowed  All

Local Group Memberships  *Administrators
Global Group memberships *None

The command completed successfully.

```

```
Select Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\CSLab>whoami
cybersecurity\csclab

C:\Users\CSLab>whoami /groups

GROUP INFORMATION
-----
Group Name                                     Type                SID                Attributes
-----
Everyone                                     Well-known group    S-1-1-0            Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Local account and member of Administrators group Well-known group    S-1-5-114          Group used for deny only
BUILTIN\Administrators                     Alias               S-1-5-32-544       Group used for deny only
BUILTIN\Users                             Alias               S-1-5-32-545       Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\INTERACTIVE                   Well-known group    S-1-5-4            Mandatory group, Enabled by default, Enabled group
CONSOLE LOGON                             Well-known group    S-1-2-1            Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Authenticated Users           Well-known group    S-1-5-11           Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\This Organization              Well-known group    S-1-5-15           Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Local account                 Well-known group    S-1-5-113          Mandatory group, Enabled by default, Enabled group
LOCAL                                     Well-known group    S-1-2-0            Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\NTLM Authentication            Well-known group    S-1-5-64-10        Mandatory group, Enabled by default, Enabled group
Mandatory Label\Medium Mandatory Level     Label               S-1-16-8192

C:\Users\CSLab>net user

User accounts for \\CYBERSECURITYLA
-----
Administrator      CSLab      DefaultAccount
Guest               LabAdmin   Student1
Student2            WDAGUtilityAccount

The command completed successfully.

C:\Users\CSLab>net user LabAdmin

User name           LabAdmin
Full Name
Comment
User's comment
Country/region code 000 (System Default)
Account active       Yes
Account expires      Never
Password last set    8/31/2026 3:42:37 AM
Password expires     10/22/2026 3:42:37 AM
Password changeable  8/31/2026 3:42:37 AM
Password required    Yes
User may change password Yes
Workstations allowed All
Logon script
User profile
Home directory
Last logon
Logon hours allowed All
Local Group Memberships *Administrators *Users
Global Group memberships None
The command completed successfully.
```

What privileges are associated with administrative membership?

Membership in the Administrators group provides extensive control over the system. An administrator can create or remove user accounts, install software, modify system settings, change permissions, manage services, and access protected system resources. Therefore, administrator privileges should only be given when they are actually required.

Why should routine activities preferably be performed using a standard account?

Routine activities should be performed using a standard account because it follows the principle of least privilege. A standard account has limited permissions, so if malicious software is accidentally executed or an unsafe action is performed, its ability to make system-wide changes is reduced. Administrative privileges can then be used only when necessary.

What security risks arise when excessive users are granted administrator privileges?

Giving administrator privileges to too many users increases the attack surface of the system. If any privileged account is compromised, an attacker may gain extensive control over the computer, modify system settings, access protected data, install malicious software, create other accounts, or disable security controls. Limiting administrator membership reduces the potential impact of a compromised account.

6. Access-control experiments

Authentication vs. Authorisation

Authentication is the process by which the operating system establishes who is attempting to use the system in this lab, by validating a username and password against the credentials stored in the local Security Accounts Manager (SAM) database.

Authorisation is a separate, later step: once identity has been established, Windows consults the access control list (ACL) attached to the specific resource being requested, and determines what that already-authenticated identity is permitted to do with it read, write, create, delete, and so on.

A simple way to keep the two apart: authentication answers “who are you”, while authorisation answers “now that I know who you are, what are you allowed to do”. Windows performs the first step once, at logon, and the second step repeatedly, every time a token is presented to a securable object.

Directory Structure

A test directory tree was created on the system drive:

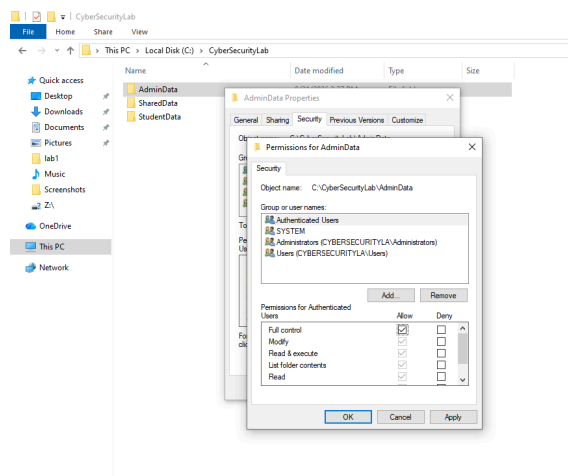
C:\CyberSecurityLab

C:\CyberSecurityLab\AdminData

C:\CyberSecurityLab\StudentData

C:\CyberSecurityLab\SharedData

A single test file was placed in each subdirectory (admin_notes.txt, student_notes.txt and shared_notes.txt respectively) to serve as the object being read, modified, created and deleted during testing



```

Select Administrator: Command Prompt
08/25/2026 03:27 PM <DIR> AdminData
08/25/2026 03:28 PM <DIR> SharedData
08/25/2026 03:27 PM <DIR> StudentData
0 File(s) 0 bytes
5 Dir(s) 18,235,731,968 bytes free

C:\CyberSecurityLab>cd AdminData
C:\CyberSecurityLab\AdminData>echo Hello World > "Adminfile.txt"
C:\CyberSecurityLab\AdminData>cd ..
C:\CyberSecurityLab>cd StudentData
C:\CyberSecurityLab\StudentData>echo Hello World > "Studentfile.txt"
C:\CyberSecurityLab\StudentData>cd ..
C:\CyberSecurityLab>cd SharedData
C:\CyberSecurityLab\SharedData>echo Hello World > "Sharedfile.txt"
C:\CyberSecurityLab\SharedData>cd..
C:\CyberSecurityLab>cd Admin
The system cannot find the path specified.
C:\CyberSecurityLab>cd AdminData

```

7. Permission matrix

A simple permission matrix can be maintained in the report:

User	AdminData	StudentData	SharedData
LabAdmin	Full Control	Full Control	Full Control
Student1	No Access	Modify	Modify
Student2	No Access	Modify	Modify

detailed record of the experiment:

User	Operation	Directory	Result
LabAdmin	Read	AdminData	Allowed
LabAdmin	Modify	AdminData	Allowed
Student1	Read	AdminData	Denied
Student1	Modify	AdminData	Denied
Student1	Create	StudentData	Allowed
Student1	Delete	StudentData	Allowed
Student2	Read	StudentData	Allowed
Student2	Modify	StudentData	Allowed
Student2	Access	AdminData	Denied

Why Does the Access Behaviour Occur?

The observed behaviour occurs because Windows checks the user's security token against the ACL of the requested file or directory. The token contains information about the user's identity and group memberships.

For example- if Student1 belongs to CyberLabUsers and StudentData grants CyberLabUsers Modify permission, Student1 receives that access through the group. On the other hand, if AdminData is restricted to LabAdmin, a standard user such as Student1 will not be able to access it.

Therefore, Authentication → identifies the user → Authorization → checks permissions → Access is allowed or denied.

This experiment demonstrates how Windows **Access Control Lists (ACLs)** can separate resources between users. Properly configured permissions help prevent unauthorized users from reading, modifying, creating, or deleting protected data. It also demonstrates the principle of least privilege, where standard users are given only the permissions required for their activities while administrative resources remain restricted. This reduces the potential impact of an accidentally or maliciously misused account.

8. UAC observations

User Account Control exists so that membership in the Administrators group does not, by itself, mean every process an account runs automatically carries full administrative power. Under UAC's Admin Approval Mode, an administrator's logon session is issued two access tokens: a filtered, standard-user-equivalent token that ordinary processes use by default, and a full administrator token that is attached to a process only after the user explicitly consents. This turns privilege elevation into a visible, one-off event rather than a standing condition of the session.

1. Standard-User Execution

A **standard user account** operates with normal, limited privileges. It can perform regular activities such as running applications, creating files, and changing settings within its permitted scope. When the user attempts an operation that requires administrative privileges, Windows normally displays a **credential prompt** asking for an administrator's username and password. Without valid administrator credentials, the operation cannot proceed.

2. Administrator Execution

An **administrator account** has the ability to perform administrative operations, but Windows does not normally give every application unrestricted administrative privileges automatically. With UAC enabled, applications generally start with a **standard user-level token**. When an operation requires administrative privileges, Windows requests **elevation through a UAC prompt**.

3. Elevated Execution

Elevated execution occurs when Windows grants an application or process the additional administrative privileges required to perform a privileged operation. For an administrator, this usually involves approving the **UAC prompt**. For a standard user, elevation generally requires entering valid administrator credentials.

Difference Between the Three

Execution	Privileges	UAC Behaviour
Standard-user execution	Limited/normal privileges	Administrative operation requires administrator credentials
Administrator execution	Administrator account, but normally runs applications with standard privileges initially	UAC asks for confirmation when elevation is required

Elevated execution	Administrative privileges are actively granted to the process	Process can perform the requested administrative operation
---------------------------	---	--

```

Select Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user3>cd..
C:\Users>cd..
C:\>cd CyberSecurityLab
C:\CyberSecurityLab>cd StudentDat
The system cannot find the path specified.
C:\CyberSecurityLab>cd StudentData
C:\CyberSecurityLab\StudentData>type Studentfile.txt
Access is denied.
C:\CyberSecurityLab\StudentData>echo "hello world by student by user 3" > Studentfile.txt
Access is denied.
C:\CyberSecurityLab\StudentData>echo "hello world by student by user 3" > StudentData.txt
C:\CyberSecurityLab\StudentData>dir
Volume in drive C has no label.
Volume Serial Number is A662-381A

Directory of C:\CyberSecurityLab\StudentData

File Not Found

```

In this above screenshot we can see that even though user3 has created the file studentData, he is not able to read it and it all comes down to one simple thing inheritance of permission by child from parent folder as we have used OI and CI, the permissions are passed down to the files inside that folder and we have only given write permission to the parent folder not the read so its sub files will also not be able to read.

1. Why is UAC required?

UAC is required to limit unnecessary use of administrative privileges. It prevents applications from automatically obtaining full administrative access simply because an administrator is logged in. When a sensitive operation requires elevation, UAC provides a confirmation or credential prompt. This helps reduce the impact of malware, accidental system changes, and unauthorized modifications.

2. Does UAC mean that an administrator account has no administrative privileges?

No. An administrator account still has administrative privileges. UAC controls when those privileges are actually used by a process.

With UAC enabled, an administrator normally works with a standard-user token for everyday applications. When administrative privileges are required, the user can approve elevation and Windows provides the process with an elevated administrative token.

Therefore:

Administrator account ≠ permanently elevated process

Instead:

Administrator account → Standard execution → UAC approval → Elevated execution

3. What security risks arise if UAC protections are weakened or disabled?

Weakening or disabling UAC reduces a layer of protection between normal application execution and administrative operations.

Potential risks include:

- Malware can make system-level changes more easily if administrative privileges are available.
- Malicious applications may install software, modify system settings, or create persistence mechanisms with less user intervention.
- Accidental administrative changes become easier to make.
- The user receives fewer warnings when sensitive operations are attempted.
- A compromised administrator session can have a greater impact on the system.

Therefore, disabling UAC increases the potential impact of malicious or accidental privileged operations.

4. How does privilege elevation relate to the Principle of Least Privilege?

The Principle of Least Privilege (PoLP) states that a user or process should receive only the privileges necessary to perform its current task.

UAC supports this principle by allowing an administrator to operate normally with limited privileges and obtain administrative privileges only when required.

For example:

Opening a text editor → normal privileges

Changing a protected system setting → UAC elevation → administrative privileges

9. Command reference table

Command Prompt (CMD) is a command-line interface provided by Windows for executing commands and performing administrative, diagnostic, file-management, networking, and system-analysis tasks.

File and Directory Management

Command	Purpose	Example Syntax	Key Options	Security Relevance
dir	Lists files and subdirectories in the current or a specified directory.	dir C:\CyberSecurityLab /a	/a- (incl. hidden/system), /s- (recurse), /o -(sort order)	Reveals hidden or system-attribute files an Explorer window may not show by default useful when auditing for concealed tools or staged data.
cd	Displays or changes the current working directory.	cd C:\CyberSecurityLab	cd .. (parent), cd \ (root)	Minimal risk directly, but correct navigation is a prerequisite for every other file-based command in a session.
mkdir / md	Creates a new directory.	mkdir C:\CyberSecurityLab\Temp	(none required)	Unrestricted directory creation inside sensitive paths (e.g. under Program Files, when permissions are misconfigured) can be abused to stage a malicious payload.
rmdir / rd	Removes a directory.	rmdir /s /q C:\CyberSecurityLab\Temp	/s (remove tree), /q (no confirmation)	/s /q run against the wrong path is one of the most common causes of accidental, unrecoverable data loss from the command line exactly the kind of destructive capability that should be limited to accounts that genuinely need it.

copy	Copies one or more files.	copy admin_notes.txt C:\CyberSecurityLab\SharedData\	/y (suppress overwrite prompt)	Can be used to relocate sensitive files into a more accessible location the copying account controls a simple but effective exfiltration technique.
move	Moves or renames files and directories.	move student_notes.txt C:\CyberSecurityLab\StudentData\	(none required)	Similar exfiltration risk to copy, with the added effect of removing the file from its original (and possibly audited) location.
del / erase	Deletes one or more files.	del /f /q file.txt	/f (force read-only), /s (recurse)	An account with only write access can use del /f /s /q to destroy logs or evidence the reason log directories are usually given restrictive, append-oriented permissions in production.
type	Displays the contents of a text file.	type C:\CyberSecurityLab\AdminData\admin_notes.txt	(none required)	The simplest possible way to exfiltrate a file's contents a reminder that read access alone, without any write access, is often sufficient to leak sensitive data.
tree	Displays the directory structure graphically.	tree C:\CyberSecurityLab /f	/f (include file names)	Useful for quickly mapping a filesystem's layout exactly why it is also a common first step for anyone (administrator or attacker) reconnoitring a host.

```

Administrator: Command Prompt
C:\Windows\system32>dir C:\CyberSecurityLab /a
Volume in drive C has no label.
Volume Serial Number is ACC0-E27F

Directory of C:\CyberSecurityLab

08/31/2026  02:36 PM  <DIR>      .
08/31/2026  02:36 PM  <DIR>      ..
08/31/2026  02:37 PM  <DIR>      AdminData
08/31/2026  02:38 PM  <DIR>      SharedData
08/31/2026  02:38 PM  <DIR>      StudentData
               0 File(s)              0 bytes
               5 Dir(s) 15,594,037,248 bytes free

C:\Windows\system32>type C:\CyberSecurityLab\AdminData\admin_notes.txt
The system cannot find the file specified.

C:\Windows\system32>type C:\CyberSecurityLab\AdminData\admin_data.txt
The system cannot find the file specified.

C:\Windows\system32>type C:\CyberSecurityLab\AdminData\admindata.txt
The system cannot find the file specified.

C:\Windows\system32>type C:\CyberSecurityLab\AdminData\admin_test.txt
Administrator test data

C:\Windows\system32>tree C:\CyberSecurityLab /f
Folder PATH listing
Volume serial number is ACC0-E27F
C:\CYBERSECURITYLAB
├── AdminData
│   └── admin_test.txt
├── SharedData
│   └── shared_test.txt
└── StudentData
    └── student_test.txt

C:\Windows\system32>^S_

```

System Information

Command	Purpose	Example Syntax	Key Options	Security Relevance / Potential Misuse
systeminfo	Displays detailed configuration: OS version, install date, BIOS version, installed hotfixes, memory.	systeminfo	/s (remote system), /fo (output format)	The installed-hotfix list reveals unpatched vulnerabilities; also a very common early reconnaissance command for both defenders auditing a machine and attackers profiling a foothold.
hostname	Prints the computer's network name.	hostname	(none)	Trivial alone, but a building block for asset inventory and for correlating log entries across a network.

ver	Displays the Windows version string.	ver	(none)	A quick way to check a machine's build number against a known-vulnerable-version list before deeper analysis.
set	Displays, sets or removes Command Prompt environment variables.	set MYVAR=test	(none)	Careless scripts sometimes store credentials or API keys in environment variables; enumerating them can reveal secrets that were never meant to persist in plaintext.
tasklist	Lists currently running processes.	tasklist /v	/v (verbose, shows owner), /svc (services per process)	A core tool for spotting an unfamiliar or suspiciously named process, and for identifying which account context a process is running under.
taskkill	Terminates a process by PID or image name.	taskkill /PID 1234 /F	/F (force), /IM (by image name)	Legitimate for stopping a hung or malicious process, but the same capability lets malware (or a rogue script) kill security software if the account running it has sufficient rights.

```

Command Prompt
TCP 10.0.2.15:51590 dns:https CLOSE_WAIT
TCP 10.0.2.15:51629 180.149.59.136:https CLOSE_WAIT
TCP 10.0.2.15:51632 server-108-159-80-95:https CLOSE_WAIT

C:\>tasklist

Image Name PID Session Name Session# Mem Usage
=====
System Idle Process 0 Services 0 8 K
System 4 Services 0 152 K
Registry 92 Services 0 26,160 K
smss.exe 328 Services 0 948 K
csrss.exe 420 Services 0 4,920 K
wininit.exe 496 Services 0 6,020 K
csrss.exe 508 Console 1 5,084 K
winlogon.exe 596 Console 1 10,728 K
services.exe 612 Services 0 8,908 K
lsass.exe 648 Services 0 17,536 K
svchost.exe 752 Services 0 28,028 K
fontdrvhost.exe 760 Services 0 3,656 K
fontdrvhost.exe 768 Console 1 4,252 K
svchost.exe 864 Services 0 14,896 K
svchost.exe 920 Services 0 7,272 K
svchost.exe 984 Services 0 5,056 K
dwm.exe 360 Console 1 73,216 K
svchost.exe 376 Services 0 9,380 K
svchost.exe 672 Services 0 5,896 K
svchost.exe 916 Services 0 15,632 K
svchost.exe 1068 Services 0 6,860 K
svchost.exe 1104 Services 0 10,496 K
svchost.exe 1216 Services 0 6,628 K
svchost.exe 1236 Services 0 12,956 K
svchost.exe 1256 Services 0 6,784 K
svchost.exe 1264 Services 0 7,484 K
svchost.exe 1300 Services 0 68,596 K
svchost.exe 1324 Services 0 5,876 K
Memory Compression 1452 Services 0 125,972 K
svchost.exe 1492 Services 0 15,128 K
svchost.exe 1512 Services 0 7,304 K
svchost.exe 1576 Services 0 10,132 K
svchost.exe 1644 Services 0 7,300 K
svchost.exe 1652 Services 0 7,960 K
svchost.exe 1824 Services 0 9,056 K
svchost.exe 1840 Services 0 11,920 K

```

Your device is at risk of being out of date. It will need to restart to install updates. Select a time to restart.

9/1/2026

Networking

Command	Purpose	Example Syntax	Key Options	Security Relevance / Potential Misuse
ipconfig	Displays IP configuration for all network adapters.	ipconfig /all	/all (full detail), /release, /renew, /flushdns	The starting point for confirming which network segment a machine is actually on including verifying the lab VM's NAT/host-only isolation is genuinely in effect.

ping	Tests reachability of a host using ICMP echo requests.	ping 8.8.8.8	-t (continuous), -n (count)	Also widely used in bulk, via a script, for network sweeps during reconnaissance one reason many hardened networks block inbound ICMP.
tracert	Traces the route packets take to a destination, hop by hop.	tracert 8.8.8.8	-d (skip DNS resolution), -h (max hops)	Can reveal internal network topology and the presence of intermediate routing or security devices to anyone able to run it.
arp	Displays or modifies the ARP cache (IP-to-MAC address mappings).	arp -a	-a (display), -d (delete entry)	Reviewing the ARP table is a basic check for ARP spoofing / man-in-the-middle activity on the local network segment.
netstat	Displays active connections and listening ports.	netstat -ano	-a (all), -n (numeric), -o (owning PID)	One of the fastest ways to spot an unexpected listening port or an outbound connection to an unfamiliar address a staple of manual malware triage.
nslookup	Queries DNS for a given hostname or record.	nslookup example.com	-type= (record type)	Can confirm whether DNS resolution has been tampered with (e.g. pointed at a rogue DNS server) a common precursor to redirection or phishing attacks.

```
Command Prompt
-x86_wcf-system.identitymodel.selectors_b03f5f7f11d50a3a_10.0.19041.1_none_2459ed2ac211ecde
-x86_wcf-system.identitymodel_b03f5f7f11d50a3a_10.0.19041.1_none_e00fcd7d17e3f70
-x86_wcf-system.identitymodel_b03f5f7f11d50a3a_10.0.19200.110_none_43ef36dd0b83d4e0
-x86_wcf-system.io.log_b03f5f7f11d50a3a_10.0.19041.1_none_9b56331c41e0b85
-x86_wcf-system.runtime.serialization_b03f5f7f11d50a3a_10.0.19041.1_none_c8294510b1b4c5d7
-x86_wcf-system.runtime.serialization_b03f5f7f11d50a3a_10.0.19200.110_none_25877e2690ba5b47
-x86_wcf-system.servicemodel_b03f5f7f11d50a3a_10.0.19041.1_none_d5e154fc7c748380
-x86_wcf-system.servicemodel_b03f5f7f11d50a3a_10.0.19200.110_none_333f8e125b7a18f9
-x86_windows-defender-management-powershell_31bf3856ad364e35_10.0.19041.1_none_3d95cd0a88523a81
-x86_wpf-penimc_31bf3856ad364e35_10.0.19200.756_none_7b197b7314960ef6
-x86_wpf-penimc_31bf3856ad364e35_10.0.19200.968_none_7b10afe7149c7e3
-x86_wpf-penimc_31bf3856ad364e35_10.0.19200.969_none_7b11b031149c013a
-x86_wpf-presentationframework_31bf3856ad364e35_10.0.19200.756_none_3deec49165df32b6
-x86_wpf-presentationframework_31bf3856ad364e35_10.0.19200.968_none_3de5f90565e57ba3
-x86_wpf-presentationframework_31bf3856ad364e35_10.0.19200.969_none_3de6f04f6e494fa
-x86_wpf-presentationhostdll_31bf3856ad364e35_10.0.19200.756_none_3f73d5c42fe806d4
-x86_wpf-presentationhostdll_31bf3856ad364e35_10.0.19200.968_none_3f6b0a382fee4fc1
-x86_wpf-presentationhostdll_31bf3856ad364e35_10.0.19200.969_none_3f6c0a822fed0918
-x86_wpf-presentationhostexe_31bf3856ad364e35_10.0.19041.1_none_c96a551071df37e
-x86_wpf-presentationhostproxy_31bf3856ad364e35_10.0.19041.1_none_845d88ac60cf32b0
-x86_wpf-presentationnative_31bf3856ad364e35_10.0.19200.912_none_2655b6b9564c3923
-x86_wpf-reachframework_31bf3856ad364e35_10.0.19200.756_none_cf9293bb38a24b71
-x86_wpf-reachframework_31bf3856ad364e35_10.0.19200.968_none_cf89c32f38a8045e
-x86_wpf-reachframework_31bf3856ad364e35_10.0.19200.969_none_cf8ac87938a7adb5
-x86_wpf-windowsbase_31bf3856ad364e35_10.0.19200.756_none_26d5756f57aeaa8c
-x86_wpf-windowsbase_31bf3856ad364e35_10.0.19200.968_none_26cca96357b53379
-x86_wpf-windowsbase_31bf3856ad364e35_10.0.19200.969_none_26cdaa2d57b44cd0
-x86_wcf-system.workflow.activities_31bf3856ad364e35_10.0.19200.101_none_01ff5e3387c37f14
-x86_wcf-system.workflow.componentmodel_31bf3856ad364e35_10.0.19200.101_none_5b7d84ca1dda7a85
-x86_wcf-system.workflow.runtime_31bf3856ad364e35_10.0.19200.101_none_34b448eaarcbcb99

C:\>netstat

Active Connections

Proto Local Address Foreign Address State
TCP 10.0.2.15:49710 4.213.25.241:https ESTABLISHED
TCP 10.0.2.15:51420 dns:https CLOSE_WAIT
TCP 10.0.2.15:51482 dns:https CLOSE_WAIT
TCP 10.0.2.15:51590 dns:https CLOSE_WAIT
TCP 10.0.2.15:51629 180.149.59.136:https CLOSE_WAIT
TCP 10.0.2.15:51632 server-180-159-80-95:https CLOSE_WAIT

C:\>
```

Q.write a command which looks normal (listed commands only) but can result into security risks.

Tasklist.-It is the cmd that look nrmal but result into security risk.

tasklist is a normal Windows command used to display currently running processes. However, when executed by an unauthorized user, it can disclose information about the system's running applications and services.

Potential security implications:

- Reveals processes and services currently running.
- May expose security software or monitoring tools.
- Helps an attacker understand the system's configuration.
- Can assist reconnaissance before attempting further attacks.

10. PowerShell observations

PowerShell is Microsoft's command-line shell and scripting environment designed for system administration and automation.The major conceptual difference from traditional CMD is that PowerShell works with objects.

Five Commands Selected for Detailed Demonstration

As required by the task, five of the eight cmdlets above were selected for closer demonstration, chosen to cover system information, process management, user administration, permissions and networking respectively:

Cmdlet	Purpose	Example Syntax
Get-ComputerInfo	Full system inventory in a single call.	Get-ComputerInfo Select-Object WindowsProductName, OsVersion, CsProcessors
Get-Process	Inspect running processes and resource usage.	Get-Process Sort-Object CPU -Descending Select-Object -First 10
Get-LocalUser	Enumerate local accounts and their status.	Get-LocalUser Select-Object Name, Enabled, LastLogon
Get-Acl	Inspect the owner and permissions on a folder.	Get-Acl "C:\CyberSecurityLab\AdminData" Format-List
Get-NetIPAddress	Inspect IP configuration per adapter.	Get-NetIPAddress Where-Object {\$_.AddressFamily -eq "IPv4"}

```

Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/powershell

PS C:\Users\CSLab> Get-ComputerInfo

WindowsBuildInfo           : 10041.1.amd64fre.rs_release.191206-1406
WindowsCurrentVersion      : 6.3
WindowsEditionId           : Core
WindowsInstallationType    : Client
WindowsInstallDateFromRegistry : 8/23/2026 2:17:59 PM
WindowsProductId           : 90131-10000-90000-AA141
WindowsProductName         : Windows 10 Home
WindowsRegisteredOrganization : 
WindowsRegisteredOwner    : Windows User
WindowsSystemRoot          : C:\Windows
WindowsVersion             : 20H2
BiosCharacteristics        : {4, 7, 15, 16...}
BiosIdVersion              : (VBOX - 1)
BiosBuildNumber            : 
BiosCaption                : Default System BIOS
BiosCodeSet                : 
BiosCurrentLanguage        : 
BiosDescription            : Default System BIOS
BiosEmbeddedControllerMajorVersion : 0
BiosEmbeddedControllerMinorVersion : 0
BiosFirmwareType          : Bios
BiosIdentificationCode     : 
BiosInstallLabelLanguages : 
BiosInstallDate            : 
BiosLanguageEdition       : 
BiosListOfLanguages        : 
BiosManufacturer          : innosetk GmbH
BiosName                   : Default System BIOS
BiosOtherFirmware         : 
BiosPrimaryBIOS           : True
BiosReleaseDate            : 12/1/2006 5:38:00 AM
BiosSerialNumber           : VirtualBox-33f3e6d7-56e8-45f7-8344-7a7af670293
BiosSMBIOSBIOSVersion      : VirtualBox
BiosSMBIOSMajorVersion     : 2
BiosSMBIOSMinorVersion    : 5
BiosSMBIOSPresent         : True
BiosSoftwareFirmwareState : Running
BiosStatus                 : OK
BiosSystemBIOSMajorVersion : 0
BiosSystemBIOSMinorVersion : 1
BiosSystemBIOSRevision     : 1
BiosVersion                : VBOX - 1
CsAdminPasswordStatus      : Unknown
CsAutomaticManagedPagefile : True
CsAutomaticResetBootOption : True
CsAutomaticResetCapability : True
CsBootOptionOnLimit        : 
CsBootOptionWakeOnS3       : 
CsBootOptionWakeOnS4       : True
CsBootOptionWakeOnS5       : True
CsBootStatus               : Normal boot
CsCaption                   : CYBERSECURITYLAB
CsChassisBootupState        : Safe
CsChassisSKUNumber         : 
CsCurrentTimezone          : XXX
CsDisplayInEffect           : 
CsDescription              : AT/AT COMPATIBLE
CsHostName                 : CyberSecurityLab
CsDomain                   : WORKGROUP
CsDomainRole                : StandaloneWorkstation
CsEnabledBiosLightSavingsTime : True
CsFrontPanelResetStatus    : Unknown
CsHyperervisorPresent      : True
CsHypervisorSupported       : False

```

```
Windows PowerShell

PS C:\Users\CSLab> Get-Process

Handles NPM(K) PM(K) WS(K) CPU(s) Id SI ProcessName
-----
120 7 1680 4748 0.70 4596 0 AggregatorHost
302 17 4996 28248 0.36 9220 0 audio
100 11 6476 12044 0.36 9220 0 audio
251 16 4236 20932 0.19 1740 1 backgroundTaskHost
282 20 36072 8044 1.99 740 1 chrome
246 22 25620 18184 8.84 1456 1 chrome
486 49 344520 34256 122.52 1548 1 chrome
184 13 3704 6560 0.69 2912 1 chrome
1704 69 116504 186996 497.13 1540 1 chrome
231 11 7032 2268 0.19 5868 1 chrome
320 26 23960 25476 41.45 5920 1 chrome
248 20 26420 17080 0.53 5916 1 chrome
392 19 17156 7524 0.31 7012 1 chrome
244 15 13384 11056 0.20 7424 1 chrome
403 17 23884 49740 142.75 8100 1 chrome
236 21 27504 49100 0.17 9308 1 chrome
232 21 21556 38840 0.13 9316 1 chrome
78 6 4824 936 0.05 4032 1 cmd
265 14 12980 8852 0.70 4852 1 conhost
277 14 7584 26420 2.66 8712 1 conhost
526 20 1184 3560 0.40 0 csrss
519 23 28012 5196 544 1 csrss
450 17 4144 8376 1.45 6326 1 ctfmon
233 17 3480 6148 0.30 3724 1 dllhost
211 16 4176 6120 0.03 3828 0 dllhost
903 46 53016 76452 396 1 dwm
2594 109 120180 111544 48.45 4856 1 explorer
200 17 18672 5016 0.30 2140 1 fileCaching
50 7 1748 3764 788 1 fontdrvhost
50 7 3512 3144 726 0 fontdrvhost
0 0 60 8 0 1 idle
1359 27 4640 12356 692 0 lsass
1165 56 36296 12656 5.27 6928 1 NDIS.sys
0 0 1148 11352 1768 0 Memory Compression
459 16 4680 18544 2036 0 NDIS.sys
164 10 2780 2420 0.34 956 1 msedge
1248 43 30692 11820 8.77 1040 1 msedge
342 18 18852 5980 2.27 6324 1 msedge
196 12 1072 4592 0.26 6340 1 msedge
301 16 11272 5692 0.39 6356 1 msedge
1179 50 40520 59160 4.52 532 1 msedgeWebView2
352 64 18944 27464 1.98 2372 1 msedgeWebView2
311 25 54848 1780 3.14 2396 1 msedgeWebView2
210 11 3860 4484 0.22 3352 1 msedgeWebView2
192 12 9536 1420 0.19 3756 1 msedgeWebView2
247 22 19220 2200 2.41 4720 1 msedgeWebView2
309 22 61160 67900 2.97 4776 1 msedgeWebView2
169 10 1732 1654 0.22 5088 1 msedgeWebView2
341 17 13832 18972 1.45 5996 1 msedgeWebView2
192 12 18164 5640 0.20 6728 1 msedgeWebView2
1146 42 30880 6112 6.42 6872 1 msedgeWebView2
329 18 12344 1004 1.48 6896 1 msedgeWebView2
854 205 32116 152456 2716 0 WebEngine
210 87 4184 3420 3476 0 WinSrv
691 40 24132 11316 10.95 3866 1 OneDrive
749 64 81144 89964 4.14 7500 1 powershell
0 8 4180 2304 92 0 Registry
238 13 2940 2284 0.44 1872 1 RuntimeBroker
294 14 3596 12188 0.41 3460 1 RuntimeBroker
314 16 5640 14832 0.46 4316 1 RuntimeBroker
366 21 7456 18580 8.66 5892 1 RuntimeBroker
576 27 12140 21856 3.50 6112 1 RuntimeBroker
155 10 2256 11716 0.43 8300 1 RuntimeBroker
707 35 11820 23844 2.82 1836 1 SearchApp
1420 185 12164 5884 8.15 7336 1 SearchApp
```

```
Windows PowerShell

PS C:\Users\CSLab> Get-LocalUser

Name Enabled Description
-----
Administrator False Built-in account for administering the computer/domain
CSLab True A user account managed by the system.
DefaultAccount False A user account managed by the system.
Guest False Built-in account for guest access to the computer/domain
lsAdmin True
lsAdmin1 True
lsAdmin2 True
WDAGUtilityAccount False A user account managed and used by the system for Windows Defender Application Guard scenarios.

PS C:\Users\CSLab> Get-NetAC
Get-NetAC : The term 'Get-NetAC' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
[Line] char(1)
Get-NetAC
+ CategoryInfo          : (ObjectNotFound: (Get-NetAC:String) [], CommandNotFound)Exception
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\CSLab> Get-NetIP

Directory: C:\Users

Path Owner Access
----
CSLab BUILTIN\Administrators NT AUTHORITY\SYSTEM Allow FullControl...

PS C:\Users\CSLab> Get-NetIPAddress

IPAddress : fe80::7240:98a0:c5af:671a%3
InterfaceIndex : 3
InterfaceAlias : Ethernet
AddressFamily : IPv6
Type : Unicast
PrefixLength : 64
PrefixOrigin : WellKnown
BufferOrigin : Link
AddressState : Preferred
ValidLifetime : Infinite ([Timespan]:MaxValue)
PreferredLifetime : Infinite ([Timespan]:MaxValue)
DhcpSource : False
PolicyStore : ActiveStore

IPAddress : fd17:625c:f037::2:3232:932e:b990:a22d
InterfaceIndex : 3
InterfaceAlias : Ethernet
AddressFamily : IPv6
Type : Unicast
PrefixLength : 64
PrefixOrigin : RouterAdvertisement
BufferOrigin : Link
AddressState : Preferred
ValidLifetime : 23:56:18
PreferredLifetime : 89:56:18
DhcpSource : False
PolicyStore : ActiveStore

IPAddress : fd17:625c:f037::2:2c76:5b16:20c4:3258
InterfaceIndex : 3
InterfaceAlias : Ethernet
AddressFamily : IPv6
Type : Unicast
```

3 PowerShell vs. Command Prompt

Aspect	Command Prompt (CMD)	PowerShell
--------	----------------------	------------

Command syntax	Individual, often terse legacy utilities with inconsistent flag conventions.	Uniform Verb-Noun cmdlet naming (Get-Process, Stop-Service), discoverable via Get-Help and Get-Command.
Output format	Plain, unstructured text intended for a human reader.	Live .NET objects with typed properties that can be filtered, sorted and reformatted without re-parsing text.
Automation	Batch (.bat) scripting: limited data structures and weak error handling.	PowerShell scripts (.ps1) support functions, modules and try/catch, and can call the full .NET framework.
Object-based processing	Commands are piped as raw text, so chaining tools often means fragile text-parsing.	Commands are piped as objects, so a property (e.g. a process's memory usage) is accessed directly without regex.
Administrative capabilities	A fixed, comparatively small set of built-in commands.	Almost every modern Windows/Azure administrative surface ships a PowerShell module, and PowerShell Remoting (WinRM) extends this to remote hosts.
Security implications	A smaller native surface area, correspondingly easier to defend and monitor exhaustively.	Greater power is double-edged: it is also the preferred post-exploitation tool for attackers with a foothold, which is exactly why Script Block Logging, Constrained Language Mode and execution policies exist.

11. Security weaknesses identified and Recommended Mitigations

Part asks for at least five weaknesses in Windows user and access management, drawn from the preceding experiments. The six below were each observed, directly or by extension, during Parts C–G, and are documented.

Weakness	Cause	Security Impact	Detection Method	Recommended Mitigation
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Excessive administrator privileges	Users are made members of the Administrators group for convenience, so that day-to-day work never triggers a UAC prompt or permission error.	Every process the account runs including a compromised browser tab or a malicious email attachment can request the full administrator token, giving malware the same reach as the human user.	net localgroup Administrators / Get-LocalGroupMember Administrators, compared against a documented list of who genuinely needs administrative rights.	Issue standard accounts for routine work and a separate administrator account used only when elevation is actually required, per Section 8.
Unnecessary or unused user accounts	Test or temporary accounts (a decommissioned StdUser, a leftover guest account) are created for a task and never removed afterward.	Each additional enabled account is another credential an attacker can target, and stale accounts are rarely monitored as closely as active ones.	net user lists every local account; cross-checking against a current roster of who should actually have access reveals orphaned entries.	Disable (net user <name> /active:no) or delete accounts as soon as their purpose ends, and periodically re-audit the account list.
Inappropriate file and folder permissions	Inheritance is left broken after a one-off change (as deliberately done to AdminData/StudentData in Section 6), or a folder is over-permissioned to “make an error go away.”	As Section 7 shows directly, a missing or overly broad ACE either blocks a legitimate user or grants an account far more access than its role requires.	icacls <path> or Get-Acl <path> against a documented baseline of intended permissions for each folder.	Apply the principle of least privilege explicitly per folder (as in Section 6.3), and review permissions whenever inheritance is deliberately broken.

Excessive or careless group membership	A user is added to a group (LabUsers, or worse, Administrators) to solve an immediate access problem, without confirming the group's full set of resulting rights.	Group membership is transitive across every resource that trusts the group, so one careless addition can silently grant access to far more than the folder the user actually needed.	net localgroup <groupname> or Get-LocalGroupMember <groupname>, checked against the group's documented purpose.	Create narrowly scoped groups per purpose (as LabUsers was scoped to SharedData-style access only) and review membership on a regular cadence.
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Additional weaknesses worth noting from the same experiments, more briefly: improper sharing permissions on SharedData if Modify were mistakenly granted to Everyone rather than the two named accounts; and a general lack of auditing, since none of the account, permission or elevation changes made in this lab were logged for later review enabling Object Access auditing on the CyberSecurityLab tree would close that gap.

13. Conclusion

This lab provided a systematic understanding of **Windows operating-system fundamentals and security mechanisms**, progressing from the Windows boot process to user identities, groups, access control, privilege elevation, and security-relevant command-line tools. The experiments demonstrated that Windows security is implemented through multiple interconnected layers rather than through a single mechanism.

The study of the **boot sequence** established how firmware, Windows Boot Manager, BCD, the Windows Boot Loader, kernel initialization, device drivers, system services, and the logon process work together to bring the system to a usable state. It also demonstrated the security importance of boot integrity and mechanisms such as Secure Boot.

The experiments with **users, groups and NTFS access control** demonstrated the distinction between authentication and authorization: authentication establishes **who the user is**, while authorization determines **what that authenticated user is permitted to do**. The use of ACLs and permission testing showed how access to files and directories can be restricted according to users and groups.

The study of **User Account Control (UAC)** further demonstrated how Windows separates normal execution from elevated execution. Administrative capability does not mean that every process should automatically operate with elevated privileges. Controlled elevation therefore provides an additional layer of protection against unnecessary use of administrative authority.

Finally, the investigation of **Command Prompt and PowerShell** showed that Windows provides extensive built-in tools for inspecting identities, group membership, processes, networking, services and access permissions. Commands such as `whoami /groups`, `net localgroup`, `icacls`, and `Get-Acl` make the relationship between users, privileges and resources observable and auditable.

The recurring principle throughout the lab was **least privilege**. Excessive administrator privileges, unnecessary accounts, excessive group membership and inappropriate file permissions all demonstrate how convenience can lead to broader access than is actually required. Effective Windows security therefore depends not only on having security mechanisms available, but on **configuring them narrowly, granting only necessary privileges, controlling elevation, and regularly reviewing the resulting access configuration**.

Overall, the lab established that effective Windows security is a combination of **appropriate identity management, carefully configured authorization, controlled privilege elevation, and continuous inspection and auditing**. These practical exercises provided a foundation for understanding how misconfiguration can weaken otherwise effective security mechanisms and why the principle of least privilege should remain central to Windows security administration.

14. References

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